LINUX COMMANDS

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OUTLINE

I. Linux commands

1. Usual commands
2. Pipe, backtick
3. System commands
4. Enviroment variables
5. Groups and users
6. File .cshrc

II. Vi editors

1. Insert mode
2. Normal mode

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USUAL COMMANDS

1.1 List files and directories

Syntax

ls [file/directory-name]

Option

-l (long)	: long listing of files and directory
-a (all)	: show all files start with .(dot)
-t (time)	: sort by modification time
-r (reverse)	: reserve order while sorting

1.2 Change current directory

Syntax

cd directory-name

Eg:

cd ..	: change to previous level hier directory
cd ../../..	: change to 3 previous level hier directory
cd –	: change to previous directory

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USUAL COMMANDS

1.3 Make directory (folder)

Syntax

mkdir [options] directory

Options

-m (mode) : set permission mode
-p (parents) : no error if existing, make parent directories as needed
-v (verbose) : print a message for each created directory

1.4 Remove the file or directory

Syntax

rm [options] [file-name/directory-name]

Options

-rf : remove all files/directory with subdirectories

1.5 Move / rename file or directory

Syntax

mv [source] [destination]

USUAL COMMANDS

1.6 Create a link

Syntax:

ln [options] [old file/dir] [new file/dir]

Options:

-s : creates a symbolic link
-v(verbose) : shows file names as they are processed

1.7 Compress file / dir

Syntax

tar [options] [tar-file-name] [source file/dir name]

Options

-c (create) : create the file
-f (filename) : name of archive file
-z (zip) : compress the file
-x (extract) : extract the file from archive

Eg:

tar -zcvf map_driver.tar.gz map_driver
tar -zxvf map_driver.tar.gz
tar -zxvf map_driver.tar.gz -C tmp/ : extract to tmp dir

USUAL COMMANDS

1.8 View a file

Syntax: brief diff

```
more        [filename]
cat         [filename]
less        [filename]
head -n [number]    [filename]
tail -n [number]    [filename]
vi          [filename]
```

1.9 Create a file

Syntax:

```
touch [filename]      : create an empty
```

1.10 Output redirect

Using the following symbols:

```
“>”         :output of commands will be stored in a file
“>>”        :output of commands will be appended to a file
```

USUAL COMMANDS

1.11 Count data

Syntax:

wc [options] filename

Option:

-l	: line in file
-w	: word in file
-c	: bytes in file

```
>>>>  wc u7800hrcarx5h_vio_vio_tx_grp1_hold.summary
35  440 3754 u7800hrcarx5h_vio_vio_tx_grp1_hold.summary
```

35 line; 440 word; 3754 byte

1.12 Sort data

Syntax:

sort [options] filename

Option:

-d	: dictionary order
-n	: numeric order
-r	: reserve order
-u	: unique
-k	: fileld

```
>>>>  cat test_1_12.txt
blue 30
red 20
orange 40
green 10
```

```
>>>>  sort -k 2 test_1_12.txt
green 10
red 20
blue 30
orange 40
```


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USUAL COMMANDS

1.13 Search patterns in file

Syntax:

grep [option] pattern [file]

Options:

-e : regular expression
-f : obtain a pattern in file
-v : invert the sense of matching, select non-matching lines
-n : line numbers matching pattern
-c : count lines matching the pattern

Eg:

grep -e VIO -e data rep.tim : search pattern “VIO” and “data” in rep.tim file
grep -f t1 rep.tim : search patterns obtained in file t1 in file rep.tim file

1.14 Search file

Syntax:

find [search dir] [options] [pattern]

Options:

-name : search file with name in pattern options
-type : search type of file

PIPE AND BACKTICK

2.1 Pipe

Syntax: using vertical bar to redirect out to another commands
Eg:

```
>>>>  cat test_1_12.txt | grep "n" | wc -l
2
```

2.2 Backtick

Usually use “`” to execute the command in backtick 1st
Eg:

```
>>>>  echo `date`
Wed Aug 27 16:06:03 JST 2025
```


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SED (edit stream of data)

3. Sed (edit stream of data)

Faster than interactive editor like vi

Syntax:

sed [option] [script]][file]

Option:

-e : add command to running command

-f : add file to running command

Eg:

```
Command >>  sed 's/VIOLATED/ANHN/' violated.line
 slack (ANHN)                -0.271
 slack (ANHN)                -0.258
 slack (ANHN)                -0.257
 slack (ANHN)                -0.191
 slack (ANHN)                -0.191


Command >>  sed 's/VIOLATED/ANHN/ ; s/slack/nguyen/' violated.line
 nguyen (ANHN)                -0.271
 nguyen (ANHN)                -0.258
 nguyen (ANHN)                -0.257
 nguyen (ANHN)                -0.191
 nguyen (ANHN)                -0.191
```

SYSTEM COMMANDS

4.1 Check the process

Syntax:

top [options]

Option:

-u : show the specific user

-c : show the command lines

Notes:

check load average, mem, swap

type F to sort other field

type h for help

Eg:

```
top - 16:12:59 up 456 days,  7:24, 934 users,  load average: 2.22, 2.43, 2.46
Tasks: 7220 total,   4 running, 7075 sleeping, 127 stopped,  14 zombie
Cpu(s):  4.3%us,  2.4%sy,  0.5%ni, 91.5%id,  0.0%wa,  0.0%hi,  1.4%si,  0.0%st
Mem:  264532252k total, 263287544k used,  1244708k free,    11868k buffers
Swap: 134217724k total, 113349636k used, 20868088k free, 12666980k cached

  PID USER      PR  NI  VIRT  RES  SHR S %CPU %MEM    TIME+  COMMAND
21066 naraiar_  20   0  200m 9.8m 3360 R 100.0  0.0  11552:32 gvim
26369 umamahpa  39  19 3414m 1.4g 2016 R 74.9  0.6  21509:08 etxproxy
26765 umamahpa  20   0  249m 8056 2396 R 61.8  0.0  17407:56 wnck-applet
26999 17431     20   0 1800m 639m 4996 S 11.5  0.2 782:37.62 etxproxy
 4122 thang_b_  20   0 18640 6692  904 R  4.1  0.0   0:00.26 top
22253 14559     20   0 3397m 2.6g 5160 S  3.5  1.0 401:30.89 etxproxy
31424 17431     20   0  4196  872  624 D  3.5  0.0   0:39.07 du
17402 17432     20   0  287m  86m 4688 S  2.2  0.0 539:51.88 gnome-terminal
```


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SYSTEM COMMANDS

4.2 Check resources

Syntax: show the space usage

du [options]

Option:

-s : summarize the

-h : show human readable format

```
>>>>  du -sh *
172K    drvFailureReason
58M     hold_MAX_LT_CW_usr
57M     hold_MIN_HT_RCW_usr
163M    mtarpt
21M     setup_MAX_HT_RCWT_usr
23M     setup_MAX_LT_CWT_usr
4.0K    splitFailureReason
0       test 1 12.txt
```

Syntax: show the memory usage

free [options]

Option:

-h : human readable format

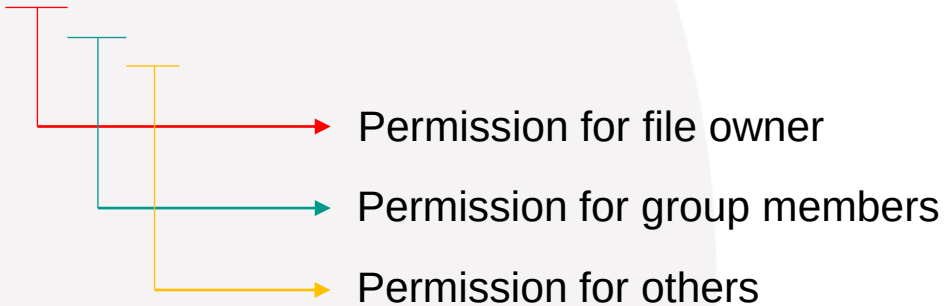
```
>>>>  free -h .
             total       used       free     shared    buffers     cached
Mem:          252G       250G       2.2G       904K        12M        10G
-/+ buffers/cache:       239G        13G
Swap:         127G       108G        19G
```


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GROUP AND USER

5.1 File permission

drwxr-xr-x



1st field “-” for file
“d” for directory
“l” for link

set of 3 character defined permission triplet
“r” for read
“w” for rite
“x” for execute

```
total 80
drwxr-xr-x  2 anhn users  4096 Aug 31 01:42 alias
-rw-r--r--  1 anhn users  1008 Aug 30 21:55 a.no_sshX
-rw-r--r--  1 anhn users  1031 Aug 30 21:58 a_sshX
```

Linux File Permission Codes

Permissions	Binary	Octal	Description
---	000	0	No permissions
--x	001	1	Execute-only permission
-w-	010	2	Write-only permission
-wx	011	3	Write and execute permissions
r--	100	4	Read-only permission
r-x	101	5	Read and execute permissions
rw-	110	6	Read and write permissions
rwx	111	7	Read, write, and execute permissions

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GROUP AND USER

5.2 Change permission:

Syntax

chmod [options] mode filename

Notes:

+	: add permission
-	: subtract permission
u	: set permission for user
g	: set permission for group
o	: set permission for other

```
 Command >>  chmod 775 t1

TuanAnh [@server4 ]$  ->/mnt/nfs/server9dat
 Command >>  ll t1
-rwxrwxr-x 1 anhn users 0 Aug 31 23:48 t1
```

```
 Command >>  chmod u-x-r t1

TuanAnh [@server4 ]$  ->/mnt/nfs/server9data/
 Command >>  ll t1
-----w---- 1 anhn users 0 Aug 31 23:48 t1
```

5.3 Mask out the permission

Syntax:

umask

Eg: umask 111 → permsssion 666

```
 Command >>  umask 111

TuanAnh [@server4 ]$  ->/mnt/nfs/server9data/
 Command >>  ll t3
-rw-rw-rw- 1 anhn users 0 Sep  1 00:39 t3
```


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ENVIRONMENT VARIABLE

6. Global environment variable

Store data in memory that can be easily accessed by any program or script running from every shell
Checked by “printenv” or “env” command
Default shell envir. variable:

. PATH:
PATH=/usr/kerberos/bin:/usr/local/bin:/bin:/usr/bin:/usr/X11R6/bin

→ 5 local dirs where shell look for commands seperated by semi-colon.

. DISPLAY: set X display name
DISPLAY=localhost:21.0

. LM_LICENSE_FILE: file for tool’s license
LM_LICENSE_FILE=5280@fileserver

set path and exmaple

```
 Command >>  env
LANG=en_US.UTF-8
USER=anhn
LOGNAME=anhn
HOME=/home/anhn
PATH=/usr/kerberos/bin:/usr/local/bin:/bin:/usr/bin:/usr/X11R6/bin
MAIL=/var/spool/mail/anhn
SHELL=/bin/tcsh
SSH_CLIENT=10.10.10.54 48163 22
SSH_CONNECTION=10.10.10.54 48163 10.10.10.24 22
SSH_TTY=/dev/pts/19
TERM=xterm
DISPLAY=localhost:21.0
HOSTTYPE=x86_64-linux
VENDOR=unknown
OSTYPE=linux
MACHTYPE=x86_64
SHLVL=1
PWD=/mnt/nfs/server9data/proj/socio_yak/DESIGN/TRAINING/anhn
GROUP=users
HOST=server4
REMOTEHOST=10.10.10.54
HOSTNAME=server4
INPUTRC=/etc/inputrc
LS_COLORS=no=00:fi=00:di=00;34:ln=00;36:pi=40;33:so=00;35:bd=40;33;01:cd
rj=00;31:*.taz=00;31:*.lzh=00;31:*.zip=00;31:*.z=00;31:*.Z=00;31:*.gz=00
G_BROKEN_FILENAMES=1
SSH_ASKPASS=/usr/libexec/openssh/gnome-ssh-askpass
LESSOPEN=|/usr/bin/lesspipe.sh %s
LM_LICENSE_FILE=5280@fileserver
```

FILE .cshrc

7. File .cshrc

Located in your home dir, /home/user/
Be re-loaded by shell when we start a new shell.
Be can define functions, commands commonly used in this file

```
set prompt="\n%{\033[34m%}TuanAnh%{\033[30m%}[%{\033[31m%}@%m %{\033[30m%}]%{\0
alias cd    'chdir \!*; set prompt="\n%{\033[34m%}TuanAnh %{\033[30m%}[%{\033[3
alias cd    'chdir \!*; set prompt="\n%{\033[34m%}TuanAnh %{\033[30m%}[%{\033[3
#alias cd 'chdir \!*; ls -lthr; set prompt = "================================

## Aliases ##
source /home/anhn/alias/commands
source /home/anhn/alias/tools
```

Alias command allow to create alias name for common commands to help keep minimum in typing

```
alias srv8  'ssh server8'
alias srv9  'ssh server9'
alias tar_cmd   'tar -zcvf'
alias utar_cmd  'tar -zxvf'
alias visor 'Visor QA2 3'
alias le    'less -S'
```

VI EDITOR

Vi is the original editor in Linux system

Syntax

vi [filename], file will be created if does not exist

2 modes

- normal mode
- insert mode

1. Insert mode:

Type “I” key

Press “Esc” key to exit mode

Exit Vi:

. q to quit if no changes have been made to the buffer data

. q! to quit and discard any changes made to the buffer data

. w filename to save the file under a different filename

. wq! or ZZ to save the buffer data to the file and quit

2. Normal mode:

2.1 Move the cursor

PageDown (or Ctl-f) to move forward one screen of data

PageUp (or Ctl-b) to move backward one screen of data

“G” to move to the last line in the buffer

“#num + G” t move to the line number num in the buffer.

“gg” to move to the first line in the buffer

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VI EDITOR

2.2 Editing commands

Commands	Description
x	Delete the character at the current cursor position.
dd	Delete the line at the current cursor (stored in buffers).
3dd	Delete 3 lines at the current cursor (stored in buffers
dw	Delete the word at the current cursor position.
d$	Delete to the end of the line from the current cursor position.
a	Append data after the current cursor position.
A	Append data to the end of the line at the current cursor position.
r char	Replace a single character at the current cursor position with char.

VI EDITOR

2.3 Common commands

Commands	Description
yy	Copy the current line
u	Undo the last change
U	Undo all the change to the line
:s/old/new/g	Replace all occurrences of old in a line
:#,#s/old/new/g	Replace all occurrences of old between two line numbers
:%s/old/new/g to	Replace all occurrences of old in the entire file
:sp “filename”	Split 2 windows parallel, using “Ctrl+w” to move cursor btw 2 windows
:vs “filename”	Split 2 windows vertically

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VI EDITOR

2.4 Set commands

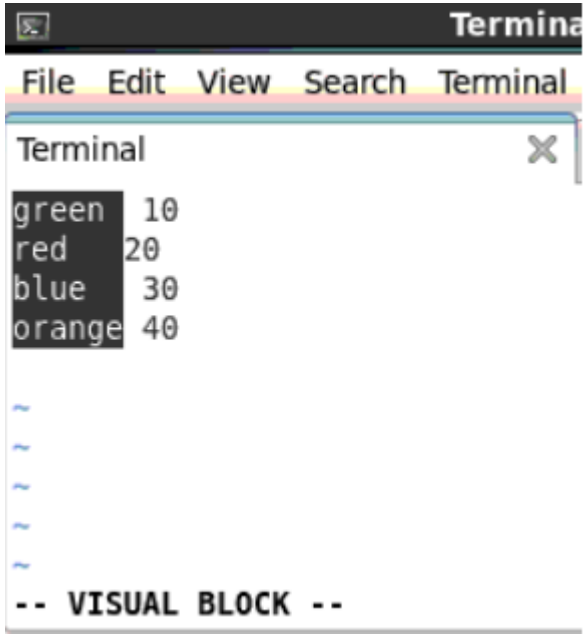
Commands	Description
: set number (: set nu)	Display number
:set nonumber (:set nonu)	No display number
: set syntax on	Set syntax highlighting and color highlighting for a file type, Syntax definition files: /usr/share/vim/vim73/syntax/
: set syntax off	No highlighting syntax
:set wrapscan (:set ws) :set nowrapscan (:set nows)	Breaks line if too long
:set tabstop=8 (: set ts=8)	Tab key displays 8 spaces

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VI EDITOR

2.5 Working with blocks

“Ctrl+V ” : to visual block mode .
“Shift+Insert” : insert text to every beginning line of selected block



REFERENCES

Linux command line and shell scripting Bible – Richard Burn

Basic vi command

https://www.cs.colostate.edu/helpdocs/vi.html

Vi cheat sheet

http://www.lagmonster.org/docs/vi.htmt

THANK YOU !